Bingjian Huang

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Dynamic Graphics Project Lab (DGP), Department of Computer Science, University of Toronto

Education

Ph.D. in Human-Computer Interaction *2020-Present*
Dynamic Graphics Project Lab (DGP), University of Toronto, Ontario, Canada
Supervisor: Prof. Daniel Wigdor

B.E. in Computer Science and Technology — GPA 3.67/4.00 *2016-2020*
Tsinghua University, Beijing, China

Minor in Finance and Entrepreneurship — GPA 3.81/4.00 *2017-2020*
Tsinghua University, Beijing, China

Publications

AeroHaptix: A Wearable Vibrotactile Feedback System for Enhancing Collision Avoidance in UAV Teleoperation
***Bingjian Huang**, Zhecheng Wang, Qilong Cheng, Siyi Ren, Hanfeng Cai, Antonio Alvarez Valdivia, Karthik Mahadevan, Daniel Wigdor*
IEEE Robotics and Automation Letters 2025

VibraForge: A Scalable Prototyping Toolkit For Creating Spatialized Vibrotactile Feedback Systems
***Bingjian Huang**, Siyi Ren, Yuewen Luo, Qilong Cheng, Hanfeng Cai, Yeqi Sang, Mauricio Sousa, Paul H Dietz, Daniel Wigdor*
CHI 2025

Investigating the Effect of Intensity and Frequency on Vibrotactile Localization Accuracy
***Bingjian Huang**, Paul H. Dietz, Daniel Wigdor*
IEEE Transactions on Haptics 2024

Cohabitant: The Design, Implementation, and Evaluation of a Virtual Reality Application for Interfaith Learning and Empathy Building
*Mohammad Rashidujjaman, Reem Ayad, Ashratuz Zavin Asha, **Bingjian Huang**, Selin Okman, Dina Sabie, Hasan Shahid Ferdous, Robert Soden, Syed Ishtiaque Ahmed*
CHI 2024

User Burden of Microinteractions: An In-lab Experiment Examining User Performance and Perceived Burden Related to In-situ Self-reporting
*Xinghui Yan, Yuxuan Li, **Bingjian Huang**, Sun Young Park, Mark W Newman*
MobileHCI 2021

Toward Lightweight In-situ Self-reporting: An Exploratory Study of Alternative Smartwatch Interface Designs in Context
*Xinghui Yan, Shriti Raj, **Bingjian Huang**, Sun Young Park, Mark W Newman*
IMWUT (UBICOMP) 2020

Designing and Evaluating Hand-to-Hand Gestures with Dual Commodity Wrist-Worn Devices
*Yiqin Lu, **Bingjian Huang**, Chun Yu, Guanhong Liu, Yuanchun Shi*
IMWUT (UBICOMP) 2020

Research Experience

Visiting PhD student *2025.1-2025.6*
Responsive Environment, MIT Media Lab, Advisor: Prof. Joseph Paradiso

- Building electrical and magnetic neural stimulation interfaces
- Developing closed-loop haptic control for robot teleoperation

Graduate Research Assistant *2020.8-Present*

Dynamic Graphics Project Lab (DGP), University of Toronto, Advisor: Prof. Daniel Wigdor

- Conducted psychophysics studies to investigate fundamental haptic perception, e.g., vibrotactile spatial acuity, phantom sensations, multisensory integration.
- Built open-source vibrotactile suit that supports over 120 vibrotactile actuators with low-latency and fine-grained control.
- Deployed haptic systems in various applications, including virtual reality, UAV teleoperation, and accessibility.

Visiting Scholar and Research Intern *2019.7-2019.9*

Interaction Ecologies Group, School of Information, University of Michigan, Ann Arbor, Advisors: Prof. Mark W. Newman, Prof. Sun Young Park

- Researched on in-situ low-burden self-reporting tools.
- Developed 11 Android smartwatch self-reporting prototypes to investigate different design concepts.
- Conducted user studies about scenarios and techniques with User Enactment Method.

Undergraduate Research Assistant *2018.3-2019.11*

Pervasive Computing Group, Tsinghua University, Adviser: Prof.Yuanchun Shi, Prof.Chun Yu

- Developed a 3D Telepresence software framework for co-presence experience.
- Implemented 3D reconstruction algorithms such as TSDF and Marching Cubes in CUDA architecture.
- Researched hand-to-hand gesture design space exploration and smartwatch recognition.
- Developed a real-time gesture recognition system using IMUs and LSTM models, validated in daily scenarios.
- Conducted 50+ interviews and user studies with participants

Skills

Programming Languages: C/C++/Python/Java

Programming Tools: Matlab/Qt/UnityVR/CUDA/PyTorch/Android Studio/Rapberry Pi

Electronic Hardware: PCB Design/CAD Design/Embedded Design

Languages: English(TOFEL 115 (speaking: 25), GRE 159+170+3.5)

Research Interests

VR/AR/MR, novel input methods and future user interfaces

Human Perception and Illusion and how they affects user experience in virtual environments

Wearable Computing with smartwatch, smart rings and smart clothes

Honors and Awards

Wolfond Fellowship ($20,000, $10,000 each year), University of Toronto *September, 2020*

GPHL Scholarship for Academic Excellence(Top 20%), Tsinghua University *September, 2019*

SOHU Scholarship for Academic Excellence(Top 10%), Tsinghua University *September, 2017*

2nd Prize, National Olympiad in Informatics in Provinces(NOIP) *November, 2014*

Miscellaneous

Sports Enthusiast: Skiing, Basketball, High Diving, Hiking